Stochastic Optimization

STAT 525 - Topics in Computational Statistics

Shulei Wang

University of Illinois at Urbana-Champaign

shuleiw@illinois.edu

https://sites.google.com/view/shulei-wang/



Overview

Stochastic Optimization

Sample Average Approximation

Stochastic Approximation

Stochastic Optimization

- We consider the following stochastic optimization problem

$$\min_{x \in \Omega} f(x) := \mathbb{E}(F(x, \xi))$$

where ξ is a random variable and $F(x, \xi)$ is convex with respect to x for any given ξ.

- If we write the distribution of ξ as $P(\cdot)$, then

$$\mathbb{E}(F(x, \xi)) = \int F(x, \xi) dP(\xi).$$

We are minimizing the expectation of a function.

Example I: Inventory Model



- A company needs to prepare x of a certain product (e.g. macbook) to satisfy demand d. The cost of ordering is c per unit, cost backorder is b and cost of holding extra units is e.
- We are trying to minimize the expected cost

$$\min_x \mathbb{E}\left(cx + b(d-x)_+ + e(x-d)_+\right)$$

- The demand d is usually a random variable.

Example II: Markowitz Model



- We want to invest different stocks to maximize our returns. Suppose w is weights of stock and r is random returns of stock.
- Then the Markowitz model is

$$\max_{w \geq 0, \sum w_i = 1} \underbrace{\mathbb{E}(w^T r)}_{\textit{Expected Return}} - \lambda \underbrace{\mathrm{Var}[w^T r]}_{\textit{Risk}}$$

- The returns of stock r is usually a random variable.

Example III: Large-scale ML Model

- Another application we are mainly interested in is large-scale statistical and machine learning model. We usually need to minimize empirical risk

$$\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} L(f(x_i), y_i)$$

where L is some loss function. This is also called finite sum problem.

- If we think j as a random index drawn from $\{1, \ldots, n\}$, then

$$\mathbb{E}\left(L(f(x_j), y_j)\right) = \frac{1}{n} \sum_{i=1}^{n} L(f(x_i), y_i)$$

This formulation is particularly useful when n is super large.

Stochastic Optimization

$$\min_{x \in \Omega} f(x) := \mathbb{E}(F(x, \xi))$$

- We may want to solve stochastic optimization by gradient based method

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - t^{(k)} \nabla f(x^{(k)}) \\ &= x^{(k)} - t^{(k)} \nabla \mathbb{E}(F(x^{(k)}, \xi)) \\ &= x^{(k)} - t^{(k)} \mathbb{E}(\nabla_x F(x^{(k)}, \xi)) \end{aligned}$$

- But
  - We might not know the distribution of ξ, e.g. the demand of customer.
  - Even we know the distribution, evaluating the expectation can be expensive, e.g. n can be large in empirical risk minimization.

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Sample Average Approximation

- A nature way to solve stochastic optimization: we generate random samples by Monte Carlo sampling

$$\xi_1, \ldots, \xi_N \sim P.$$

- Approximate $f(x)$ by its empirical version

$$f^N(x) = \frac{1}{N} \sum_{i=1}^{N} F(x, \xi_i)$$

- We then look for the optimum solution for the empirical function

$$\min_{x \in \Omega} f^N(x)$$

- This idea is called sample average approximation (SAA).

Sample Average Approximation

- How does sample average approximation perform? converge?

$$f_N^* = \min_x f^N(x) \quad \rightarrow \quad f^* = \min_x f(x)?$$

$$x_N^* = \operatorname*{argmin}_x f^N(x) \quad \rightarrow \quad x^* = \operatorname*{argmin}_x f(x)?$$

- They converge under some conditions, but not always.

A Hard Example

- Think about a simple example

$$\min_x f(x) = \mathbb{E}(\xi x) = 0$$

  where ξ follows standard normal distribution.
- In this example, every point is optimal solution and the optimal value $f^* = 0$.
- How about the sample average approximation?

$$\min_x \left(\frac{1}{N} \sum_{i=1}^{N} \xi_i \right) x$$

  SAA fails.

Convergence Analysis

- Under very mild conditions, we can know that

$$\sqrt{N}(f^N(x) - f(x)) \to N(0, \sigma^2(x))$$

  by central limit theorem.
- Under more stringent conditions, including Lipschitz continuity of $F(\cdot, \xi)$, convexity of $F(\cdot, \xi)$ and $f(x)$ has a unique solution, we can show

$$\sqrt{N}(f_N^* - f^*) \to N(0, \sigma^2(x^*))$$

- In order to achieve ϵ accuracy, we need

$$N \geq O\left(\frac{1}{\epsilon^2}\right)$$

Pros and Cons

- (Pros) SAA is general and can work with any deterministic algorithm. We can apply SAA on almost any problem.
- (Cons) In order to achieve ϵ accuracy, we usually need to choose N in advance. But it is difficult to choose a good one in practice.
- (Cons) The evaluation of gradient in SAA can be expensive when N is large
- (Cons) SAA cannot work with online data, i.e. the ξ_i comes sequentially.

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Stochastic Approximation

- Stochastic approximation (SA) is another popular method to solve stochastic optimization problem.

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla_x F(x^{(k)}, \xi^{(k)})$$

  Here, the stochastic gradient is unbiased, i.e. $\mathbb{E}(\nabla_x F(x, \xi^{(k)})) = \nabla f(x)$.
- The idea of stochastic approximation is first introduced by Robbins and Monro in 1951. Nowadays, it is also called stochastic gradient descent (SGD) method.
- This method has a long history, with variants having been invented and reinvented by different communities, under such names as “back propagation”, “online learning”, and “randomized Kaczmarz method”.

Example: Large-scale ML Model

$$\min_x \frac{1}{n} \sum_{i=1}^{n} f_i(x)$$

- At each iteration, we randomly choose a sample $i^{(k)}$ and then

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f_{i^{(k)}}(x^{(k)})$$

- SGD is particular efficient at the beginning.

The Difference from Gradient Descent

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla_x F(x^{(k)}, \xi^{(k)}) \quad \text{vs} \quad x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f(x^{(k)})$$

- Although they look similar, but we need completely different strategy to choose step size.
- Consider a simple example: we start from the the minimizer, i.e., $x^{(0)} = x^*$. GD will stay in the minimizer no matter how we choose step size. But SGD will work around so we need to choose step size carefully.

Key Assumption in Convergence Analysis

$$\mathbb{E}\left(\|\nabla_x F(x,\xi)\|^2\right) \leq L^2\|x-x^*\|^2 + B^2$$

- We need to assume the noise level is well controlled.
- One way to interpret the above bound is $\mathbb{E}(X^2) = \mathbb{E}(X)^2 + \mathrm{Var}(X)$. The first term controls $\|\nabla f(x)\|^2$ and the second term controls $\mathbb{E}(\|\nabla_x F(x,\xi) - \nabla f(x)\|^2)$. But sometimes $\mathbb{E}(\|\nabla_x F(x,\xi) - \nabla f(x)\|^2) \leq L^2\|x-x^*\|^2$.
- The choice of step size relies on L and B.

Key Assumption: $L = 0$ and $B \neq 0$

- In logistic regression, the loss function is

$$\begin{aligned}\min_{\beta} f(\beta) &= \frac{1}{n}\sum_{i=1}^{n} \log\left(1+\exp(-Y_i\beta^T X_i)\right) \\ &= \mathbb{E}_i\left[\log\left(1+\exp(-Y_i\beta^T X_i)\right)\right].\end{aligned}$$

- The stochastic gradient is

$$\nabla_\beta F(\beta, i) = -\frac{\exp(-Y_i\beta^T X_i)}{1+\exp(-Y_i\beta^T X_i)} Y_i X_i,$$

so $L = 0$ and $B = \max_i \|X_i\|^2$.

Key Assumption: $L \neq 0$ and $B = 0$

- In lease square problem, the loss function is

$$\begin{aligned}\min_{\beta} f(\beta) &= \frac{1}{n}\sum_{i=1}^{n}\left(\beta^T X_i - Y_i\right)^2 \\ &= \mathbb{E}_i\left[\left((\beta-\beta^*)^T X_i\right)^2\right].\end{aligned}$$

  Here we assume there exists β^* such that $(\beta^*)^T X_i = Y_i$.
- The stochastic gradient is

$$\nabla_\beta F(\beta, i) = 2(\beta-\beta^*)^T X_i X_i^T$$

  so $L = 2\mathbb{E}_i(\|X_i\|^4)^{1/2}$ and $B = 0$.

Key Assumption: $L \neq 0$ and $B \neq 0$

- If we only observe a noisy version of gradient

$$\nabla_x F(x, \xi) = \nabla f(x) + \xi,$$

where $\mathbb{E}(\xi) = 0$ and $\mathbb{E}(\|\xi\|^2) = d\sigma^2$.

- In such case,

$$\mathbb{E}\left(\|\nabla_x F(x, \xi)\|^2\right) = \|\nabla f(x)\|^2 + d\sigma^2,$$

so $L = M$ and $B = \sqrt{d}\sigma$ when f is M-smooth.

Convergence Analysis

- If we assume
  1. $\mathbb{E}\left(\|\nabla_x F(x,\xi)\|^2\right) \leq L^2\|x - x^*\|^2 + B^2$
  2. $\xi_1, \ldots, \xi_n$ are i.i.d.
  3. f is convex

  we have

$$A^{(k+1)} \leq \left(1 + (t^{(k)})^2 L^2\right) A^{(k)} - 2t^{(k)}\mathbb{E}\left((x^{(k)} - x^*)^T \nabla f(x^{(k)})\right) + (t^{(k)})^2 B^2,$$

  where

$$A^{(k)} = \mathbb{E}\left(\|x^{(k)} - x^*\|^2\right).$$

Proof:

$$\begin{aligned}
&\|x^{(k+1)} - x^*\|^2 \\
=&\|x^{(k)} - x^*\|^2 - 2t^{(k)}(x^{(k)} - x^*)^T \nabla_x F(x^{(k)}, \xi^{(k)}) + t_{(k)}^2 \|\nabla_x F(x^{(k)}, \xi^{(k)})\|^2
\end{aligned}$$

Since $x^{(k)}$ is a function of $\xi^{(1)}, \ldots, \xi^{(k-1)}$, then an application of tower property of conditional expectation suggests

$$\begin{aligned}
&\mathbb{E}((x^{(k)} - x^*)^T \nabla_x F(x^{(k)}, \xi^{(k)})) \\
=&\mathbb{E}(\mathbb{E}((x^{(k)} - x^*)^T \nabla_x F(x^{(k)}, \xi^{(k)}) | \xi^{(1)}, \ldots, \xi^{(k-1)})) \\
=&\mathbb{E}((x^{(k)} - x^*)^T \mathbb{E}(\nabla_x F(x^{(k)}, \xi^{(k)}) | \xi^{(1)}, \ldots, \xi^{(k-1)})) \\
=&\mathbb{E}((x^{(k)} - x^*)^T \nabla f(x^{(k)}))
\end{aligned}$$

Proof: Similarly, we have

$$\begin{aligned}
&\mathbb{E}(\|\nabla_x F(x^{(k)}, \xi^{(k)})\|^2) \\
=&\mathbb{E}\left(\mathbb{E}(\|\nabla_x F(x^{(k)}, \xi^{(k)})\|^2 | \xi^{(1)}, \ldots, \xi^{(k-1)})\right) \\
\leq&\mathbb{E}\left(L^2\|x^{(k)} - x^*\|^2 + B^2\right)
\end{aligned}$$

Putting everything together to have

$$A^{(k+1)} \leq \left(1 + (t^{(k)})^2 L^2\right) A^{(k)} - 2t^{(k)}\mathbb{E}\left((x^{(k)} - x^*)^T \nabla f(x^{(k)})\right) + (t^{(k)})^2 B^2.$$

Convergence Analysis: $L = 0$ and $B \neq 0$

- If we assume
  1. $\mathbb{E}\left(\|\nabla_x F(x,\xi)\|^2\right) \leq L^2\|x - x^*\|^2 + B^2$ with $L = 0$ and $B \neq 0$
  2. $\xi_1, \ldots, \xi_n$ are i.i.d.
  3. f is convex

  we have

$$A^{(k+1)} \leq A^{(k)} - 2t^{(k)}\mathbb{E}\left((x^{(k)} - x^*)^T \nabla f(x^{(k)})\right) + (t^{(k)})^2 B^2$$

  and

$$\mathbb{E}\left(f(\bar{x}^{(k)}) - f^*\right) \leq \frac{\|x^{(0)} - x^*\|^2 + B^2 \sum_{i=1}^k (t^{(i)})^2}{2\sum_{i=1}^k t^{(i)}}.$$

  Here, $\bar{x}^{(k)} = \sum_{i=1}^k t^{(i)} x^{(i)} / \sum_{i=1}^k t^{(i)}$.
- We can choose $t^{(k)} \asymp k^{-1/2}$.

Convergence Analysis: $L = 0$ and $B \neq 0$

Proof: By the convexity and assumption 1, we have

$$
\begin{aligned}
&\mathbb{E}\left(f(\bar{x}^{(k)}) - f^*\right) \\
\leq &\mathbb{E}\left(\frac{\sum_{i=1}^{k} t^{(i)}(f(x^{(i)}) - f^*)}{\sum_{i=1}^{k} t^{(i)}}\right) \\
\leq &\frac{\sum_{i=1}^{k} t^{(i)}\mathbb{E}(\nabla f(x^{(i)})^T(x^{(i)} - x^*))}{\sum_{i=1}^{k} t^{(i)}} \\
\leq &\frac{\sum_{i=1}^{k}(A^{(i)} - A^{(i+1)} + (t^{(i)})^2 B^2)}{2\sum_{i=1}^{k} t^{(i)}} \\
\leq &\frac{A^{(0)} + B^2\sum_{i=1}^{k}(t^{(i)})^2}{2\sum_{i=1}^{k} t^{(i)}}
\end{aligned}
$$

Convergence Analysis: $L \neq 0$ and $B = 0$

- If we assume
  1. $\mathbb{E}\left(\|\nabla_x F(x,\xi)\|^2\right) \leq L^2\|x - x^*\|^2 + B^2$ with $L \neq 0$ and $B = 0$
  2. $\xi_1, \ldots, \xi_n$ are i.i.d.
  3. f is m-strongly convex
  4. $t^{(k)} = m/L^2$

  we have

$$A^{(k+1)} \leq \left(1 - 2t^{(k)}m + (t^{(k)})^2 L^2\right) A^{(k)}$$

  and

$$\mathbb{E}\left(\|x^{(k)} - x^*\|^2\right) \leq \left(1 - \frac{m^2}{L^2}\right)^k \|x^{(0)} - x^*\|^2.$$

Convergence Analysis: $L \neq 0$ and $B = 0$

Proof: Because f is m-strongly convex,

$$(x - x^*)^T \nabla f(x) \geq m\|x - x^*\|^2.$$

So

$$A^{(k+1)} \leq \left(1 - 2t^{(k)}m + (t^{(k)})^2 L^2\right) A^{(k)}$$

We can complete the proof by choosing $t^{(k)} = m/L^2$.

Convergence Analysis: $L \neq 0$ and $B \neq 0$

- If we assume
  1. $\mathbb{E}\left(\|\nabla_x F(x,\xi)\|^2\right) \leq L^2\|x - x^*\|^2 + B^2$
  2. $\xi_1, \ldots, \xi_n$ are i.i.d.
  3. f is m-strongly convex

  we have

$$A^{(k+1)} \leq \left(1 - 2t^{(k)}m + (t^{(k)})^2L^2\right) A^{(k)} + (t^{(k)})^2B^2$$

- If we further choose $t^{(k)} = t \in (0, 2m/L^2)$, we have

$$\mathbb{E}\left(\|x^{(k)} - x^*\|^2\right) \leq \left(1 - 2tm + t^2L^2\right)^k \|x^{(0)} - x^*\|^2 + \frac{tB^2}{2m - tL^2}.$$

- If we further choose $t^{(k)} = 1/(L^2/m + 2mk)$, we have

$$\mathbb{E}\left(\|x^{(k)} - x^*\|^2\right) \leq \frac{c_0B^2}{L^2/m + 2mk}.$$

Convergence Analysis: Constant Step Size

$$\mathbb{E}\left(\|x^{(k)} - x^*\|^2\right) \leq \left(1 - 2tm + t^2L^2\right)^k \|x^{(0)} - x^*\|^2 + \frac{tB^2}{2m - tL^2}.$$

- Two stages:
  - Linear converge at the beginning, which can be very fast
  - Enter into a neighborhood of x^* and then variation in gradient prevents further improvement
- The radius of neighborhood is related to the step size. Smaller step size $\Rightarrow$ smaller neighborhood.

Optimality of SGD

- There is also lower bound for stochastic first-order method.
- SGD is optimal when σ^2 (B^2) is large and ϵ is small (the second term dominates the lower bound).

AGD ($\sigma^2 = 0$)	SGD	Lower Bound
Convex	Convex	Convex
$O\left(\frac{1}{\sqrt{\epsilon}}\right)$	$O\left(\frac{\sigma^2}{\epsilon^2}\right)$	$O\left(\frac{1}{\sqrt{\epsilon}} + \frac{\sigma^2}{\epsilon^2}\right)$
Strongly convex	Strongly convex	Strongly convex
$O\left(\sqrt{\kappa}\log\frac{1}{\epsilon}\right)$	$O\left(\frac{\sigma^2}{\epsilon}\right)$	$O\left(\sqrt{\kappa}\log\frac{1}{\epsilon} + \frac{\sigma^2}{\epsilon}\right)$

SGD vs GD

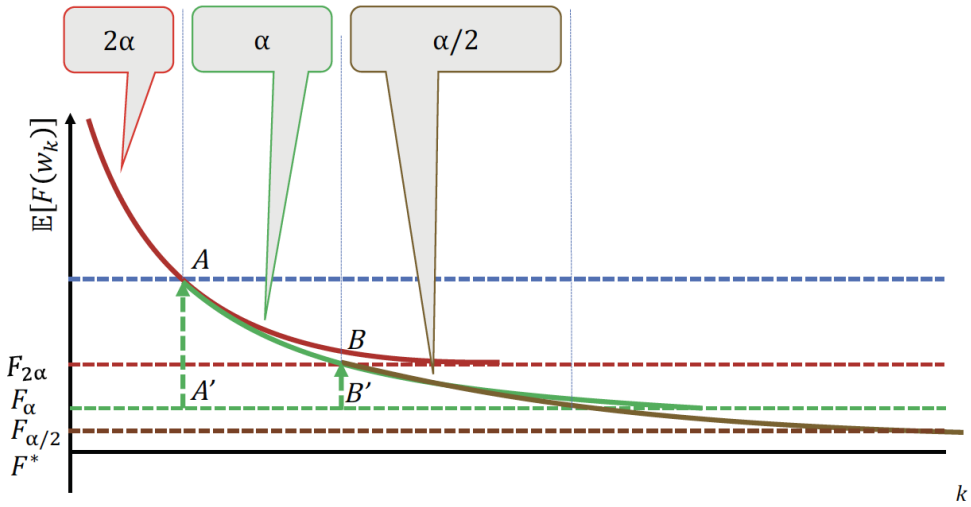
- We compare SGD and GD on empirical risk minimization (strongly convex) with n samples

	# iteration	per-iteration	total cost
GD	$\log(1/\epsilon)$	n	$n\log(1/\epsilon)$
SGD	$1/\epsilon$	1	$1/\epsilon$

- SGD is more efficient when n is large and ϵ is not too small, i.e.

$$\frac{1}{\epsilon} < n\log(1/\epsilon)$$

Implementation: Epochs



- Epoch is an important concept in stochastic gradient. We can choose fixed step size in each epoch and choose to change step size between epochs.
- In practice, Run SGD with fixed stepsizes; whenever progress stalls, reduce step size by a factor γ and continue SGD.

Implementation: Minibatch

$$\min_x \frac{1}{n} \sum_{i=1}^{n} f_i(x)$$

- At each iteration, we randomly choose a subset $S^{(k)} \subset \{1, \ldots, n\}$ and then
$$x^{(k+1)} = x^{(k)} - t^{(k)} \frac{1}{|S^{(k)}|} \sum_{i \in S^{(k)}} \nabla f_i(x^{(k)})$$
- Minibatch can help reduce variance of gradient estimate and get the algorithm more stable. But batch size is also a hyperparameter.

Implementation: Using Momentum

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla_x F(x^{(k)}, \xi^{(k)}) + \beta^{(k)} (x^{(k)} - x^{(k-1)})$$

- The idea of momentum comes from heavy-ball method and Nesterov's method.
- The theoretical gain might be mild, but it works very well in practice.